



Jayanth Kothapalli

Roll No.: 22MAE0013

Integrated MSc

Mathematics

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GitHub Profile

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Portfolio YouTube

EDUCATION

•National Institute of Technology, Warangal

2022-2027

Integrated MSc in Mathematics (Minor in CSE)

CGPA: 8.93/10.00

•Sri Chaitanya College, Khammam

2020-2022

Intermediate

MARKS: 982/1000

•SRM High School, Kodad

2020

10th

CGPA: 10/10

EXPERIENCE

•Xelron AI

May 2026 – July 2026

Software Development Intern

Remote

- Developed AI training data by analyzing GitHub repositories and closed pull requests to create repository/PR definitions, prompts, edge cases, and acceptance criteria for LLM evaluation using Cursor
- Built backend features for a FastAPI application by developing CRUD APIs, writing SQL queries and database migrations, and collaborating with the backend team using Git, GitHub, and VS Code.

•Fixity Learning Pad

May 2024 – Aug 2024

Data Scientist

Remote

- Engineered AI-powered assignment generator with multiple question formats, improving teacher efficiency by 90%
- Developed Streamlit app for 3D model feedback, achieving 95% accuracy and 80% reduction in review time
- Implemented Prompt Caching and Batch processing and reduced the AI Evaluation Cost by 83.33%

PERSONAL PROJECTS

•GenAI RAG Application

LangChain RAG Pipeline

- Implemented langchain RAG flow, starting from Document Loaders, Text splitters, Vector Databases like FAISS, Chat LLM like ChatMistralAI and Runnables like Sequential, Parallel and Passthrough.
- Deployed the application in Streamlit and explained the project to the university students.

•JaySummBot Text Summarizer

Docker, code modularization, CI/CD, AWS EC2

- Engineered text summarization pipeline using Google Pegasus model, achieving 85% ROUGE score
- Orchestrated deployment on AWS EC2 using Docker, reducing deployment time by 65%
- Established automated CI/CD pipeline with GitHub Actions and created a 1hr long YouTube video explaining project components, demonstrating end-to-end flow with deployment.

•Maximizing Taxi Fare

Data Analysis, Statistical Analysis, Hypothesis Testing

- Analyzed 2.7M+ NYC taxi trips to check if payment method (card vs cash) affects fare amount, using a t-test in Python (pandas, scipy) — found card payments average \$ 1.35 more per trip and confirmed it wasn't just random chance ($p < 0.001$).
- Built a linear regression model to predict taxi fare from trip duration, getting an R^2 of 0.76 and RMSE of \$2.85; also cleaned and prepped a raw dataset of 6.4M rows down to a usable 2.7M for analysis.

TECHNICAL SKILLS

Programming: Python, SQL, C++, Object-Oriented Programming, Data Structures and Algorithms (DSA)

Machine Learning: Scikit-learn, Random Forest, XGBoost, Clustering, Real-Coded and Binary-Coded Genetic Algorithms

Deep Learning: TensorFlow, CNNs, RNNs, Transformers

AI/MLOps: MLflow, Docker, AWS (EC2, ECR), GCP (Cloud Functions, Cloud Tasks, Buckets), Git, CI/CD

Generative AI: LangChain, RAG, LLMs, OpenAI, Anthropic (Function calling, Prompt Caching, Batch Processing)

Data Science: NumPy, Pandas, Matplotlib, Seaborn, Statistical Analysis, A/B Testing

CERTIFICATIONS

•Azure AI-900 Fundamentals – Microsoft

•Complete AI & Machine Learning Bootcamp – Udemy

•Machine Learning with Python Training Completion – Coincent